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Systems and methods for universal reference source creation and accurate secure matching

Abstract

The present disclosure is directed to systems and methods for reference source matching. Specifically, the systems and methods disclosed enable matching among tokens using a reference source. In one example, a Consolidation Platform may receive tokens from a customer environment and tokens from a reference source environment. The customer tokens may be compared to each other using AB matching. If a match does not occur, the customer tokens may further be compared to the reference source tokens via transitive matching. If a match does occur, then the customer tokens may be denoted as a match. In further example aspects, the reference source may be a universal reference token repository that comprises unique tokens. If, after a match is indicated, the matched token(s) may be compared to the universal reference token repository. If the matched token(s) does not exist, it may be added to the repository for future use.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) (1) This present application is a continuation of U.S. patent application Ser. No. 18/331,788, filed Jun. 8, 2023, entitled “SYSTEMS AND METHODS FOR UNIVERSAL REFERENCE SOURCE CREATION AND ACCURATE SECURE MATCHING,” which is a continuation of U.S. patent application Ser. No. 17/103,751, filed Nov. 24, 2020, entitled “SYSTEMS AND METHODS FOR UNIVERSAL REFERENCE SOURCE CREATION AND ACCURATE SECURE MATCHING,” which is related to the Applicant's co-pending applications: U.S. patent application Ser. No. 16/844,927, entitled “CONTEXT DRIVEN DATA PROFILING”; U.S. patent application Ser. No. 16/776,293, entitled “SYSTEMS AND METHOD OF CONTEXTUAL DATA MASKING FOR PRIVATE AND SECURE DATA LINKAGE”; and U.S. patent application Ser. No. 17/103,720, entitled “SYSTEMS AND METHODS FOR DATA ENRICHMENT.” The forgoing applications are hereby incorporated by reference herein in their entireties.

TECHNICAL FIELD

(1) The present disclosure relates to accuracy by reference source matching.

BACKGROUND

(2) Entities maintain large amounts of data that may be disorganized and/or incomplete. For example, an entity may maintain more than one incomplete record related to a subject, such as an individual, product, organization, etc.. One record may contain a subject's address, email, gender, and geographic location, while the other record may contain a subject's name, address, phone number, date-of-birth, and credit card information. Each of these records may be incomplete. Similar incomplete records may exist for products and organizations. Currently, entities desiring to reconcile these disjointed records typically must manually combine these records, which results in inefficient and time-consuming processes, as well as potential exposure to personally identifiable information.

(3) Linking, de-duplicating, or finding matches between data assets remains one of the biggest problems of most organizations due to the breadth of data sources. Multi-channel data collection and in-house replication for serving the need of customers and business stakeholders poses challenges of reconciliation. Other problems like incompleteness, data entry mistakes, and changes over time, make it difficult for establishing a master (i.e., a “golden”) record of data.

(4) Another issue that entities face is ensuring data integrity of the records they may possess. For example, an entity may have two incomplete records seemingly associated with the same data subject. However, one record may list a different email address or phone number than the other. Such data discrepancies decrease the integrity of data records and make it more difficult for an entity to reconcile multiple, incomplete data records because the entity may be unsure which record is actually correct. Furthermore, an entity may be unsure to what extent a certain record may be correct. Modern day enterprises are hampered when it comes to accurate data collection and reconciliation.

(5) As such, there is an increased need for systems and methods that can address the challenges of modern-day data collection and reconciliation, including the inefficiency of matching multiple incomplete records to the same data subject, the loss of integrity when records are inconsistent, and the potential exposure of personally identifiable information (PII), sensitive business information, and/or any form of confidential information, when such attempts to match and reconcile data records are made.

(6) It is with respect to these and other general considerations that the aspects disclosed herein have been made. Also, although relatively specific problems may be discussed, it should be understood that the examples should not be limited to solving the specific problems identified in the background or elsewhere in the disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Non-limiting and non-exhaustive examples are described with reference to the following figures.
- (2) FIG. 1 illustrates an example of a distributed system for Reference Source matching, as described herein.
- (3) FIG. 2 illustrates an example method for creating a Reference Token Repository and matching tokens using a Reference Token Repository, as described herein.
- (4) FIG. 3 illustrates an example of a distributed system that includes the Customer environment and the Consolidation Platform environment.
- (5) FIG. 4 illustrates an example of transitive matching using a Reference Source.
- (6) FIG. 5 illustrates example client sources and reference sources.
- (7) FIG. 6 illustrates an example of a distributed system that includes a Customer environment, a Consolidation environment, and a Reference Source environment.
- (8) FIG. 7 illustrates an example universal reference token set created through reference source matching.
- (9) FIG. 8 illustrates an example of a distributed system that includes a Customer environment, a Consolidation environment with a Universal Reference Token Repository, and a Reference Source environment.
- (10) FIG. 9 illustrates an example input processor for implementing systems and methods for consolidating and enriching data, as described herein.
- (11) FIG. 10 illustrates one example of a suitable operating environment in which one or more of the present embodiments may be implemented.

DETAILED DESCRIPTION

- (12) Various aspects of the disclosure are described more fully below with reference to the accompanying drawings, which form a part hereof, and which show specific exemplary aspects. However, different aspects of the disclosure may be implemented in many different forms and should not be construed as limited to the aspects set forth herein; rather, these aspects are provided so that this disclosure will be thorough and complete, and will fully convey the scope of the aspects to those skilled in the art. Aspects may be practiced as methods, systems, or devices. Accordingly, aspects may take the form of a hardware implementation, an entirely software implementation or an implementation combining software and hardware aspects. The following detailed description is, therefore, not to be taken in a limiting sense.
- (13) Embodiments of the present application are directed at systems and methods associated with data matching, specifically secure matching via a universal reference token record and intelligent source creation. Data matching processes can be used to generate insights into data that may increase data quality. For example, a data insight can include duplicate or multiple instances of data attributes within a domain and across domains, including a percentage overlap. Specifically, two data records may be associated with the same data subject. Data profiling may indicate that these two records overlap to a certain extent and may also highlight any similarities or differences in the records. Furthermore, data profiling processes may include data insights, which may include a data quality report from normalization and standardization (i.e., what percent is standard versus non-standard) or trends based on label processing (e.g., records with the same home address).
- (14) Data masking may refer to the hashing and/or encryption client-specific information. Through this application, “client” may refer to the Customer and/or the Reference Source. In order for an entity to transmit sensitive data (e.g., PII) to a third-party (e.g., Consolidation Platform), the data usually should be tokenized. In some examples described here, client-specific information may be first hashed and encrypted and then uploaded to a secure server. The tokenized information may be

retrieved from the secure server and analyzed. The tokenized client-specific information (i.e., “client tokens”) may be tokenized using a combination of hash (one way compression function) and/or encrypted, wherein the hash code may include an additional computer-generated Secure Hash Algorithm (SHA2 512/256) salt. Other hashing and encryption methods may be used, including but not limited to, SHA2 348, SHA3, TripleDES, Twofish encryption algorithms, Blowfish encryption algorithms, Advanced Encryption Standard (AEs), IDEA encryption algorithm, MD5 encryption algorithms, and/or RSA security encryption algorithms. When the third-party server receives information transmitted from a Customer or Reference Source, the information is in the form of tokens. In other words, the raw data from the Customer and/or Reference Source (i.e., “client”) is hashed and encrypted prior to being transmitted to the third-party secure server. The Consolidation Platform does not receive or process raw data.

(15) In some example aspects, a first label included in a listing of anonymized labels that corresponds to a type of information in a tokenized attributed based on an identified classifier may be included. The label can provide an anonymized identifier of a type of information represented in the attribute. The label can be generated based on any of the attribute and the classifier. For example, if an attribute relates to a name, the corresponding label can be “La1.” In these embodiments, only entities with access to a listing of the information corresponding to the labels can identify the type of information identified by each label, thereby anonymizing the data. To the third-party Consolidation Platform, the actual, unencrypted attribute or classifier may be unknown due to the tokenization of the data. For example, a certain attribute may refer to a data subject's address and have a label of “Address,” but once labeled, this label may be token “La1” to the third-party Consolidation Platform. Further description of this process can be found in U.S. patent application Ser. No. 16/844,927, which is incorporated by reference in its entirety.

(16) Data consolidating refers to consolidating two or more data records to create a single data record. For instance, a data record may include the data attributes of Name, DOB, Gender, and Email address. Another data record may include the data attributes of Name, Gender, Home address, and Phone number. If these two records are consolidated, then a single data record may be created that includes the data attributes of Name, DOB, Gender, Email address, Home address, and Phone number. Once consolidated, the consolidated data record may be known as a “composite.”

(17) A data composite may be useful to entities that may have multiple data records associated with a single data subject. For example, a bank may have a data record for a data subject's checking account and a data record for that same data subject's home loan. Both of these data records may include overlapping information, but one data record may include certain data attributes that the other data record does not include. As such, creating a composite of these data records may be beneficial to the entity, resulting in greater efficiency and higher levels of data integrity.

(18) As described herein, an entity may utilize the systems and methods presented to match certain tokens using a reference token repository. For instance, a client may have two databases with two separate records associated with the same entity (e.g., person, place, product, thing, etc.). The client may submit these records to a data masking application that produces private and secure tokens for each of these records. These tokens may then be transmitted to a consolidation platform, where the tokens are analyzed for matching. The tokens may be compared to each other in some instances (i.e., A-B matching). In other instances, the tokens may be compared to a reference token repository to identify matching records (i.e., transitive matching). If the tokens match, then one of the tokens (i.e., a unique token) may be stored in the reference token repository for future use. Such a process of storing unique tokens ensures an efficient repository because no duplicate tokens will exist in the repository.

(19) The systems and methods presented herein utilize Reference Sources to corroborate data records and fill in potential missing data attributes for a data record. A Reference Source is an entity that has data that is used as a reference point. For example, a Reference Source may be a “trusted” source, like a credit bureau, a bank, and/or a government entity. However, because “trust”

is subjective, for purposes of this application, a Reference Source is any source that is used as a reference for comparison against another source (e.g., Customer source). For example, a Customer may have multiple data records for a data subject, but some data attributes may be missing. The Customer may hash and encrypt this data (i.e., transform the raw data into tokens) and then upload the tokens to a secure server where a third-party Consolidation Platform may access the Customer tokens. Additionally, a Reference Source may have complete and verified records (i.e., “trusted” records) that are hashed and encrypted (i.e., tokenized) and then uploaded to a secure server managed by the Consolidation Platform. The third-party Consolidation Platform may access the Reference Source tokens. The third-party Consolidation Platform may then identify overlapping and/or unique tokens in the Customer data (i.e., payload) by matching the Customer tokens to the Reference Source tokens located in a Reference Source Token Repository.

(20) In some aspects, Customer tokens and Reference Source tokens may be received by the system. The Customer tokens and Reference Source tokens may be analyzed and compared. The results of the analysis and comparison may overlap in the Customer tokens and/or matching characteristics via comparison with a Reference Token Repository. Specifically, the third-party Consolidation Platform enables the identification of duplicate records and consolidation of data, but the actual consolidating of the data (e.g., merging two records associated with the same entity to create a single record) occurs in the Customer Environment.

(21) Accordingly, the present disclosure provides a plurality of technical benefits including but not limited to: enabling more efficient use of electronic resources for data matching, consolidating, and enabling data enrichment; providing more efficient storage management, since the matching, consolidating, and enrichment of a dataset may occur at a single third-party server; decreasing potential exposure of sensitive data, since all data transmitted to the third party servers is tokenized (i.e., hashed and encrypted); and decreasing manual overload of electronic devices, since the data matching, consolidating, and enrichment occur via computing devices running intelligent algorithms to identify deficiencies in a dataset and remedying those deficiencies from a Reference Source dataset, among other examples.

(22) FIG. 1 illustrates an example of a distributed system for Reference Source matching, as described herein. Example system **100** presented is a combination of interdependent components that interact to form an integrated whole for consolidating and enriching data. Components of the systems may be hardware components or software implemented on, and/or executed by, hardware components of the systems. For example, system **100** comprises client devices **102**, **104**, and **106**, local databases **110**, **112**, and **114**, network(s) **108**, and server devices **116**, **118**, and/or **120**.

(23) Client devices **102**, **104**, and **106** may be configured to receive and transmit data. For example, client devices **102**, **104**, and **106** may contain client-specific data. Client devices may download a third-party tokenization software program via network(s) **108** that may be applied to the client-specific data. The client-specific data may be stored in local databases **110**, **112**, and **114**. Once tokenized, the client-specific data is transformed into “tokens,” and these tokens may be transmitted via network(s) **108** and/or satellite **122** to server(s) **116**, **118**, and/or **120**. Server(s) **116**, **118**, and/or **120** may be third-party servers owned by a Consolidation Platform. In other examples, client-specific data may be stored in servers (in addition to or instead of local client devices and local databases) and may be tokenized and then transmitted from client servers to third-party servers via network(s) **108** and/or satellite **122**.

(24) In aspects, a client device, such as client devices **102**, **104**, and **106**, may have access to one or more datasets or data sources and/or databases comprising client-specific data. In other aspects, client devices **102**, **104**, and **106**, may be equipped to receive broadband and/or satellite signals carrying client-specific tokens or unencrypted data (or a mixture of both). The signals and information that client devices **102**, **104**, and **106** may receive may be transmitted from satellite **122**. Satellite **122** may also be configured to communicate with network(s) **108**, in addition to being able to communicate directly with client devices **102**, **104**, and **106**. In some examples, a

client device may be a mobile phone, a laptop computer, a tablet, a smart home device, a desk phone, and a wearable (e.g., a smart watch), among other devices.

(25) To further elaborate on the network topology, client devices **102**, **104**, and/or **106** (along with their corresponding local databases **110**, **112**, and **114**) may be owned by a Reference Source. The client devices **102**, **104**, and/or **106** may download a third-party software program to tokenize the data. The tokens from the Reference Source may be stored locally in local databases **110**, **112**, and/or **114**. In other examples, the tokens may be stored in remote databases/servers **116**, **118**, and/or **120**. In other examples, the tokens may be harbored in both local and external databases. After tokenization, the Reference Source may transmit data via client device(s) **102**, **104**, and/or **106** that are configured to communicate with local databases **110**, **112**, **114** and servers **116**, **118**, and **120**. The tokens from the Reference Source may be transmitted via network(s) **108** and/or satellite **122**. The Reference Source tokens may be received by third-party servers.

(26) The procedure for tokenizing and transmitting data from the Customer-side and the Reference Source-side may be similar, in that the data may be stored locally initially and subsequently hashed and encrypted on the Customer-owned and/or Reference Source-owned client devices and, once in tokenized form, finally transmitted to third-party servers for analysis, consolidation, and enrichment, among other actions. In other words, FIG. **1** depicts a network topology that may be used in a Customer Environment and/or Reference Source Environment (i.e., client devices **102**, **104**, and/or **106** may belong to the Client Environment in one example and belong to the Reference Source Environment in another example).

(27) FIG. **2** illustrates an example method for creating a Reference Token Repository and matching tokens using a Reference Token Repository, as described herein. Each token may be comprise of a single attribute or multiple attributes. For instance, a single token may refer to a group of five attributes, or in some example aspects, the token may refer to a part of a single attribute. Tokens are not limited to one-to-one matching with attributes. Method **200** begins with steps **202** and **204**, which is to receive tokens from a first source and a second source. For example, tokens may be received by a single client from two different databases (i.e., two different sources). The tokens may or may not be associated with the same entity.

(28) Prior to the transmission of the tokens at steps **202** and **204** the Customer data may be tokenized at the Customer-side. Prior to receiving the Customer data, Customer client devices may have used a tokenization software program to hash and encrypt (or “mask”) the raw Customer data prior to transmission, so what is actually transmitted to the Consolidation Platform are Customer “tokens.” Once tokenized, the Customer tokens may be transmitted and received by the third-party Consolidation Platform at step **204**. Note that receiving Customer tokens at step **204** and receiving Reference Source tokens at step **208** may be implemented simultaneously or in vice versa order (with reception of the Reference Source tokens happening before the reception of the Customer tokens).

(29) The Customer tokens may be received by the Consolidation Platform and stored on secured servers. After reception of the Customer tokens, the method proceeds to step **206**, where the tokens may be compared to each other. This comparison may utilize a matching algorithm, such as A-B matching, deterministic matching, probabilistic matching, and/or a combination of matching algorithms.

(30) After the comparison step in **206**, the tokens may then be compared to a Reference Token Repository, if one exists, at step **208**. If a Reference Token Repository exists, then the customer tokens can be further compared to the Reference Token Repository to identify tokens that may have already been stored in the repository. For example, if the comparison results at step **206** result in an affirmative match of the two customer tokens, then the tokens can be compared to the reference token repository at step **208**. Then, at step **210**, if the customer tokens are not located in the Reference Token Repository, one of the matching customer tokens (i.e., a single unique token) can be identified and then subsequently stored in the Reference Token Repository for future use at step

212. In another example, if the customer tokens do not yield a match at step **206** (e.g., through A-B matching), then the customer tokens may still be matched up using transitive matching with the Reference Token Repository at step **208**. If the customer token from the first source matches to a token in the Reference Token Repository, and the customer token from the second source matches to the same token in the Reference Token Repository, then by virtue of the transitive matching property, the customer tokens from the first source and the second source match.

(31) Upon identifying that the tokens match, at least one composite (i.e., a consolidation of at least two Customer token records) may be indicated. A composite refers to the consolidation of multiple token records to create a single token record. For example, the Customer tokens may make up multiple token records associated with the same data subject (although, the third-party Consolidation Platform is unable to identify this data subject because the information is tokenized on the third-party Consolidation Platform). A composite of the data subject may be created by consolidating the multiple token records associated with that same data subject. Specifically, one token record may include tokens that, when untokenized, represent data attributes such as Name, date of birth (DOB), Gender, and physical address, while another token record may include tokens that represent data attributes such as Name, email address, and age. The composite record, when transmitted back to the Customer and/or Reference Source environment and untokenized in the Customer and/or Reference Source environment, the consolidated record would include all of the aforementioned data attributes: Name, DOB, Gender, physical address, email address, and age. The end result is a composite record, which is a single record associated with a certain data subject and/or entity. Such a composite record is created in the Customer Environment, so while the third-party Consolidation Platform may facilitate the identification of certain records to be consolidated and/or enriched, the actual consolidation and enrichment happens at the Customer Environment prior to transmission of the Customer tokens to the Consolidation Platform. As used herein, “data subject” may be a person, product, business, entity, etc. The systems and methods described herein are not limited to simply personal data related to human data subjects but can be applied to supply chain data, product data, business entity data, etc.

(32) To further expand on step **208**, if a Reference Token Repository has not yet been created, then the method **200** may proceed from step **206** to step **210**. The identification of the unique tokens from the customer may be the initial unique tokens that comprise the Reference Token Repository. In other words, this is how a Reference Token Repository can be created. The appending of new and unique tokens to the Reference Token Repository occurs at step **212**. This step can also be the initial step in creating the Reference Token Repository. Once the Reference Token Repository is created, then step **208** can be utilized in the method as a further means of comparison and matching customer tokens more accurately using transitive matching. In some aspects, the results of the comparison between the source tokens and the reference source tokens may not yield new unique tokens. In such scenarios, the Reference Token Repository will remain unchanged.

(33) To illustrate the power of transitive matching, consider the situation where a two customer records are associated with the same data subject, but the two records do not have enough overlap/similarities for there to be successful match. This may result in a false negative if record A and record B are indeed records associated with the same data subject. To remedy this false negative problem, the system may rely upon the Reference Token Repository. By using the Reference Token Repository, the system may use a technique known as transitive closure to accurately determine if record A and record B (or, after tokenization, token A and token B) are associated with the same data subject, and if so, create a composite of those records. Token A may be matched up with the Reference Source token, and Token B may be matched up with the same Reference Source token. If the comparison of the tokens reaches a certain matching threshold that confidently determines that token A matches with the Reference Source token and that token B matches with the Reference Source token (i.e., $A=R$ and $B=R$), then we know that records A and B must match due to the transitive property.

(34) The Reference Source tokens that may be used to help generate a composite of the Customer tokens may derive from different sources. For example, a Reference Token Repository may be created purely based on customer tokens. In other instances, a Reference Token Repository may be created by one or more third-party trusted sources. In yet other examples, the Reference Token Repository may be created through a combination of customer tokens and trusted source tokens.

(35) Reference Source tokens may be received by the Consolidation Platform in the same manner as the Customer tokens. As noted earlier, the Reference Source tokens may be received prior to, simultaneously with, or after the Consolidation Platform receives the Customer tokens at steps **202** and **204**. Similar to the receive tokens steps at steps **202** and **204**, the Reference Source data may also be tokenized prior to transmission to the Consolidation Platform. The Reference Source devices may use a third-party tokenization software program (e.g., a tokenization software from the Consolidation Platform or another third-party) to tokenize its data prior to transmission. Once tokenized, the Reference Source tokens may be transmitted and received by the Consolidation Platform. The Consolidation Platform will now possess both Customer tokens and Reference Source tokens. As noted with respect to FIG. **1**, these tokens may be stored on secured servers owned by the Consolidation Platform, third-party cloud servers, and/or local devices managed by the Consolidation Platform.

(36) In some example aspects, tokens in the Reference Token Repository may possess a rank and/or confidence value based on the comparison frequency of the token to the Reference Token Repository. For example, a certain customer token that is compared to the Reference Token Repository for a first time may indicate to the system that the customer token is not in the Reference Token Repository. As such, the customer token is a “unique” customer token and should be stored in the Reference Token Repository. Subsequently, another customer token from a different data source is compared to the Reference Token Repository, and the comparison results show an affirmative match to the customer token in the Reference Token Repository. As such, that customer token is noted as a duplicate of an already-stored token in the Reference Token Repository, but the fact that the system saw another corroborating customer token that matched with a token in the Reference Token Repository indicates that the particular customer token in the Reference Token Repository has a higher likelihood of accuracy. The more duplicate tokens that are compared to the Reference Token Repository, the higher the confidence value of that certain token in the Reference Token Repository may be. Specifically, a confidence value may be indicative of comparison frequency, i.e., a confidence value of “8” indicates that the token has been compared 8 times to a duplicate token. A confidence value of “0” or “NA” indicates that a particular token stored in the Reference Token Repository has not been compared to a duplicate record, so there are no known corroborating records in the system.

(37) FIG. **3** illustrates an example of a distributed system that includes the Customer environment and the Consolidation Platform environment. The distributed system in FIG. **3** is an example data registry **300**. The data registry **300** may be comprised of the Customer environment **302** and the Consolidation Platform **304**. As shown in the Customer environment **302**, data may be received from multiple Customer databases (e.g., databases Source A and/or Source B). The data may contain multiple data attributes associated with a data subject, such as Name, Address, Phone, Email, SSN, DOB, Gender, etc. The Customer data may be tokenized inside of the Client Environment **302**. Once tokenized, the client tokens may be transmitted to the Consolidation Platform **304**.

(38) The Consolidation Platform **304** may receive the tokens from the Customer Environment **302** in the form of private and secure tokens. The Consolidation Platform **304** may receive the tokens from the Customer environment in the form of private and secure tokens. Within the Consolidation Platform **306**, the tokens from the Customer may be compared, e.g., using A-B matching techniques. At the comparison step, linked pairs (i.e., “overlap”) may be identified with tokens. In some examples, a record may also consist of a bit string (i.e., string of 1's and 0's). A “1” may

indicate that a certain data attribute exists, whereas a “0” may indicate that the data attribute does not exist. When the tokens for a record are compared, if token or tokens of records have the same value then a linked pair is established based on an established threshold. If the bit values in a matched record pair are different, then at least one of the token records has the data attribute, whereas the other token record does not. If both token records have a “0” in the column, then that particular corresponding data attribute is missing. Specifically, a linked pair may be created by the customer tokens that are transmitted to the Consolidation Platform **304**. The customer tokens may then be compared to at least one reference source token. The results of a match between the customer tokens and reference source tokens may prompt the customer to assign a new record ID to reflect the existence of a match.

(39) FIG. 4 illustrates an example of transitive matching using a Reference Source. As illustrated in example **500**, data records A (**402**) and B (**404**) may contain certain data attributes. Record A (**402**) may contain an address, email, gender, and license type. Record B (**404**) may contain name, address, phone, DOB, gender, and credit card number. When these two records are compared to create a “composite” record, however, the system may determine that there is not enough information/overlap between these two records to confidently determine that the records are related to the same data subject. In other words, comparing just record A and record B may result in a “false negative.”

(40) By introducing a Reference Source data record **406**, records A and B can be matched via the transitive property. In other words, Record A may be compared to Record R, and sufficient overlap may be present to confidently determine that the records are associated with the same data subject. Record B may also be compared to Record R, and sufficient overlap may be present to confidently determine that the records are associated with the same data subject. As such, if Record A=Record R and Record B=Record R, then Record A=Record B.

(41) The transitive matching example **400** may be applied when identifying duplicate records and/or tokens when comparing the tokens to each other and/or to a Reference Token Repository, such as in step **208** of method **200**. At least one Reference Source may be utilized to determine whether certain Customer data records are associated with the same data subject.

(42) Of note, at the Consolidation Platform, the transitive matching occurs via tokens—the data attributes displayed in FIG. 4 are for understanding purposes only, as the “data” (i.e., payload) that is actually received and processed by the Consolidation Platform is tokenized. In other words, the Consolidation Platform is blind to the underlying raw data that the tokens represent. The Consolidation Platform may or may not know which token are matched, but regardless, the Consolidation Platform does not know the content of what is matched. The Consolidation Platform, for example, does not know actual content/data of Customer data record A and Customer data record B matching Addresses. Rather, the Consolidation Platform knows that Customer token record A and Customer token record B have a matching token between the two token records (if they do indeed match), but the Consolidation Platform does not know that the matched content a token represents.

(43) FIG. 5 illustrates example client sources and reference sources. Client sources **502** may include various client records coming from different data sources, such as data source A, B, and X. Such client records may be tokenized and transmitted to the Consolidation Platform where the client records may be compared amongst each other to identify overlapping tokens. The unique tokens may be stored in a Reference Token Repository.

(44) In other examples, external reference sources may be utilized to contribute to the Reference Token Repository. For instance, tokens from a reference source may be compared to tokens from a client source, and if a match occurs, the unique tokens may be saved in a Reference Token Repository. In further examples, the client tokens and the reference source tokens may be compared to the Reference Token Repository. Based on transitive matching properties, the token may already be stored in the Reference Token Repository, which means that a unique token(s) has already

existed.

(45) FIG. 6 illustrates an example of a distributed system that includes a Customer environment, a Consolidation environment, and a Reference Source environment. Similar to FIG. 3, FIG. 6 includes a customer environment **602** and a consolidation platform **606**, as well as a reference source environment **604**. As noted previously, the customer environment may comprise data from different sources (e.g., source A, source B, etc.). These data records may be masked in the customer environment so that they are in the form of private and secure tokens. The tokens are then transmitted to the consolidation platform **606** for processing. Similarly, the reference source environment may also comprise data records. These data records may be masked according to the same algorithm as the customer environment data records. The output of the reference source data records is also private and secure tokens. These tokens may be similarly transmitted to the consolidation platform **606**.

(46) At the consolidation platform **606**, the tokens from the customer environment and the tokens from the reference source environment may be compared using known matching techniques, such as A-B or transitive closure matching. If overlap occurs, linked pairs can be created showing which bits overlap from the token. If enough overlap is indicated, then the tokens may be considered a match.

(47) FIG. 7 illustrates an example universal reference token set created through A-B matching and/or reference source matching. A universal reference set may be a composite of multiple datasets (or “token sets” at the Consolidation Platform) from Reference Sources. The universal reference set may be utilized by a Consolidation Platform to efficiently identify which data attributes from a Customer dataset may be corroborated and/or populated by a Reference Source and which Reference Source may possess the relevant data. By utilizing a universal reference set, a Consolidation Platform may more easily provide recommendations to the Customer as to which Reference Source may have the most relevant data to corroborate and/or populate certain data attributes.

(48) As illustrated, a universal token reference set can be created from a variety of client and external reference sources and/or a combination of both. For instance, two client tokens may be compared and matched via AB Matching at **702**. A single unique token record may be created based on this matching and stored in the universal token reference set. In another example, client tokens may be compared to a reference source token at **704**. Based on the transitive matching property, a single unique token record may be created and stored in the universal token reference set. In yet another example, two tokens from two reference sources may be compared via AB Matching at **706**. If a match is identified, then a single unique token record may be created and stored in the universal token reference set **710**.

(49) As illustrated, the Token Validation (text example) **708** displays the various data attributes associated with a data subject that have been compiled by various data sources (e.g., Source 1, Source 2, Source N, T Source 1, T source 2, O Source, etc.). However, in the Consolidation Platform environment (**606** from FIG. 6), the Token Validation set **708** will appear as the Universal Token Reference Set (Actual) **710**, where particular data attributes and sources are not actually identifiable (i.e., source agnostic), since they are tokenized. In other words, the Token Validation (text example) **708** displays an example of the underlying un-tokenized data that may make up a “universal” reference set, and Universal Token Reference Set (Actual) **710** is the actual, tokenized “universal” reference set that is applied during the comparison step with tokenized Customer data.

(50) FIG. 8 illustrates an example of a distributed system that includes a Customer environment, a Consolidation environment with a Universal Reference Token Repository, and a Reference Source environment. As previously illustrated in FIGS. 3 and 6, the distributed system **800** of FIG. 8 contains customer environment **802**, reference source environment **804**, and consolidation platform **806**. In addition to these components, the consolidation platform **806** also comprises a Universal Reference Token Repository **810**. The Universal Reference Token Repository **810** is comprised of

unique tokens (i.e., no duplicates. The unique tokens are identified during the data matching process. For instance, private and secure tokens from the customer environment (e.g., from source A and source B) may be transmitted to consolidation platform **806**. Similarly, private and secure tokens from the reference source environment **804** may be transmitted to the consolidation platform **806**. At the consolidation platform **806**, the tokens may be compared with one another via AB Matching and/or transitive matching and/or other matching algorithms. The results of the matching are linked pairs that show which tokens overlap and which tokens do not overlap. A certain overlapping percentage may indicate the existence of a match. In this instance, a token **812A** from the reference source environment **804** may be matched with tokens from the customer environment that come from sources A and B. Based on the linked pairs, the system indicates that the token **812A** matches with the customer tokens. As such, the Universal Reference Token Repository **810** will store a single copy of that token **812B** for future use if the comparison yields new information (i.e., tokens **812B** will be updated based on new matching information). In other words, even though the same token may derive from the customer environment and the reference source environment, the Universal Reference Token Repository only stores a single instance of this token to avoid duplication. Tokens **812A** may be a token repository, and Tokens **812B** may be a token repository. Token repository **812A** may be the same as token repository **812B** for reference source matching.

(51) FIG. **9** illustrates an example input processor for implementing systems and methods for consolidating and enriching data, as described herein. Input processor **900** may be embedded within a client device (e.g., client devices **102**, **104**, and/or **106**), remote web server device (e.g., devices **116**, **118**, and/or **120**), and other devices capable of implementing systems and methods for consolidating and enriching data. The input processing system contains one or more data processors and is capable of executing algorithms, software routines, and/or instructions based on processing data provided by at least one client source and/or Reference Source. The input processing system can be a factory-fitted system or an add-on unit to a particular device. Furthermore, the input processing system can be a general-purpose computer or a dedicated, special-purpose computer. No limitations are imposed on the location of the input processing system relative to a client or remote web server device, etc. According to embodiments shown in FIG. **9**, the disclosed system can include memory **905**, one or more processors **910**, communications module **915**, matching module **920**, and token identification module **925**. Other embodiments of the present technology may include some, all, or none of these modules and components, along with other modules, applications, data, and/or components. Still yet, some embodiments may incorporate two or more of these modules and components into a single module and/or associate a portion of the functionality of one or more of these modules with a different module.

(52) Memory **905** can store instructions for running one or more applications or modules on processor(s) **910**. For example, memory **905** could be used in one or more embodiments to house all or some of the instructions needed to execute the functionality of matching module **920** and/or token identification module **925**, as well as communications module **915**. Generally, memory **905** can include any device, mechanism, or populated data structure used for storing information. In accordance with some embodiments of the present disclosures, memory **905** can encompass, but is not limited to, any type of volatile memory, nonvolatile memory, and dynamic memory. For example, memory **905** can be random access memory, memory storage devices, optical memory devices, magnetic media, floppy disks, magnetic tapes, hard drives, SIMMs, SDRAM, RDRAM, DDR, RAM, SODIMMs, EPROMs, EEPROMs, compact discs, DVDs, and/or the like. In accordance with some embodiments, memory **905** may include one or more disk drives, flash drives, one or more databases, one or more tables, one or more files, local cache memories, processor cache memories, relational databases, flat databases, and/or the like. In addition, those of ordinary skill in the art will appreciate many additional devices and techniques for storing

information that can be used as memory **905**.

(53) Communications module **915** is associated with sending/receiving information (e.g., matched by matching module **920**, and/or identified by token identification module **925**), commands received via client devices or server devices, other client devices, remote web servers, etc. These communications can employ any suitable type of technology, such as Bluetooth, WiFi, WiMax, cellular (e.g., 5G), single hop communication, multi-hop communication, Dedicated Short Range Communications (DSRC), or a proprietary communication protocol. In some embodiments, communications module **915** sends information output by matching module **920** (e.g., matching tokens), and/or token identification module **925** (e.g., identifying a unique token based on the match to store in a Universal Reference Token Repository) to client devices **102**, **104**, and/or **106**, as well as memory **905** to be stored for future use. In some examples, communications modules may be constructed on the HTTP protocol through a secure REST server(s) using RESTful services.

(54) Matching module **920** is configured to receive at least two sets of data, e.g., dataset A and dataset B. The module **920** may also receive and maintain at least one reference dataset (e.g., Reference Source dataset, universal reference set, etc.). Matching module **920** may be configured to compare dataset A and dataset B via matching algorithms, such as AB matching. Module **920** may further be configured to compare dataset A to the reference dataset and compare dataset B to the reference dataset. The comparison analysis may reveal that dataset A matches sufficiently to the reference dataset and that dataset B matches sufficiently to the reference dataset. If dataset A matches to the reference dataset and dataset B matches to the reference dataset, then dataset A matches to dataset B via the transitive property. Matching module **920** may also accept single data records and compare the data records to data records from the at least one dataset that is maintained by transitive matching module **920**. Matching module **920** is also configured to execute the transitive matching analysis on tokens. In other words, at least two token sets may be compared against a Reference Source token set, and if token set A matches to the Reference token set and token set B matches to the Reference token set, then token set A matches token set B via the transitive property.

(55) Token Identification module **925** may be configured to identify a single unique token record that is determined to be a match to another token based on the results of matching module **920**. If a match is indicated from matching module **920**, then a copy of the matched token may be obtained by token identification module **925** and stored in a reference token repository, such as Universal Reference Token Repository **810** from FIG. 8. Token identification module **925** may also be configured to identify already-stored tokens in a token repository. For instance, based on the results of matching module **920**, a unique token may be identified but not stored in the Universal Reference Token Repository because the unique token already exists, and the Universal Reference Token Repository will not accept duplicate tokens.

(56) FIG. 10 illustrates one example of a suitable operating environment in which one or more of the present embodiments may be implemented. This is only one example of a suitable operating environment and is not intended to suggest any limitation as to the scope of use or functionality. Other well-known computing systems, environments, and/or configurations that may be suitable for use include, but are not limited to, personal computers, server computers, hand-held or laptop devices, multiprocessor systems, microprocessor-based systems, programmable consumer electronics such as smart phones, network PCs, minicomputers, mainframe computers, distributed computing environments that include any of the above systems or devices, and the like.

(57) In its most basic configuration, operating environment **1000** typically includes at least one processing unit **1002** and memory **1004**. Depending on the exact configuration and type of computing device, memory **1004** (storing, among other things, information related to detected devices, association information, personal gateway settings, and instructions to perform the methods disclosed herein) may be volatile (such as RAM), nonvolatile (such as ROM, flash

memory, etc.), or some combination of the two. This most basic configuration is illustrated in FIG. **10** by dashed line **1006**. Further, environment **1000** may also include storage devices (removable, **1008**, and/or non-removable, **1010**) including, but not limited to, magnetic or optical disks or tape. Similarly, environment **1000** may also have input device(s) **1014** such as keyboard, mouse, pen, voice input, etc. and/or output device(s) **1016** such as a display, speakers, printer, etc. Also included in the environment may be one or more communication connections, **1012**, such as LAN, WAN, point to point, etc.

(58) Operating environment **1000** typically includes at least some form of computer readable media. Computer readable media can be any available media that can be accessed by processing unit **1002** or other devices comprising the operating environment. By way of example, and not limitation, computer readable media may comprise computer storage media and communication media. Computer storage media includes volatile and nonvolatile, removable and non-removable media implemented in any method or technology for storage of information such as computer readable instructions, data structures, program modules or other data. Computer storage media includes, RAM, ROM, EEPROM, flash memory or other memory technology, CD-ROM, digital versatile disks (DVD) or other optical storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or any other tangible medium which can be used to store the desired information. Computer storage media does not include communication media.

(59) Communication media embodies non-transitory computer readable instructions, data structures, program modules, or other data. Computer readable instructions may be transported in a modulated data signal such as a carrier wave or other transport mechanism and includes any information delivery media. The term “modulated data signal” means a signal that has one or more of its characteristics set or changed in such a manner as to encode information in the signal. By way of example, and not limitation, communication media includes wired media such as a wired network or direct-wired connection, and wireless media such as acoustic, RF, infrared and other wireless media. Combinations of the any of the above should also be included within the scope of computer readable media.

(60) The operating environment **1000** may be a single computer operating in a networked environment using logical connections to one or more remote computers. The remote computer may be a personal computer, a server, a router, a network PC, a peer device or other common network node, and typically includes many or all of the elements described above as well as others not so mentioned. The logical connections may include any method supported by available communications media. Such networking environments are commonplace in offices, enterprise-wide computer networks, intranets and the Internet.

(61) Aspects of the present disclosure, for example, are described above with reference to block diagrams and/or operational illustrations of methods, systems, and computer program products according to aspects of the disclosure. The functions/acts noted in the blocks may occur out of the order as shown in any flowchart. For example, two blocks shown in succession may in fact be executed substantially concurrently or the blocks may sometimes be executed in the reverse order, depending upon the functionality/acts involved.

(62) The description and illustration of one or more aspects provided in this application are not intended to limit or restrict the scope of the disclosure as claimed in any way. The aspects, examples, and details provided in this application are considered sufficient to convey possession and enable others to make and use the best mode of claimed disclosure. The claimed disclosure should not be construed as being limited to any aspect, example, or detail provided in this application. Regardless of whether shown and described in combination or separately, the various features (both structural and methodological) are intended to be selectively included or omitted to produce an embodiment with a particular set of features. Having been provided with the description and illustration of the present application, one skilled in the art may envision variations, modifications, and alternate aspects falling within the spirit of the broader aspects of the general

inventive concept embodied in this application that do not depart from the broader scope of the claimed disclosure.

(63) From the foregoing, it will be appreciated that specific embodiments of the invention have been described herein for purposes of illustration, but that various modifications may be made without deviating from the scope of the invention. Accordingly, the invention is not limited except as by the appended claims.

Claims

1. A non-transitory, computer-readable storage medium comprising instructions recorded thereon, wherein the instructions, when executed by at least one data processor of a system, cause the system to: obtain a first token and a second token, wherein the first token includes first information that is data masked, wherein the second token includes second information that is data masked, wherein the first information is different from the second information; compare the first token and the second token to determine whether the first token and the second token match; upon determining that the first token and the second token do not match: obtain a third token from a database storing unique tokens, wherein the third token includes third information that is data masked, wherein the third token is unique in the database; without accessing the first information in human-readable form and the third information in human-readable form, compare the first token and the third token by comparing the first information that is data masked and the third information that is data masked to obtain a first indication of whether the first token and the third token match; without accessing the second information in human-readable form and the third information in human-readable form, compare the second token and the third token by comparing the second information that is data masked and the third information that is data masked to obtain a second indication of whether the second token and the third token match; upon determining that the first token and the third token match and that the second token and the third token match: determine that the first token and the second token match; and reduce memory footprint associated with the database by storing in the database a composite token including the first information and the second information, wherein the first information and the second information are data masked.
2. The non-transitory, computer-readable storage medium of claim 1, wherein instructions to store in the database an indication that the composite token is created: determine whether the composite token is included in the database; upon determining that the composite token is not included in the database, add the composite token to the database; and upon determining that the composite token is included in the database, increase a confidence value associated with the composite token, wherein the confidence value indicates an accuracy associated with the composite token.
3. The non-transitory, computer-readable storage medium of claim 1, comprising instructions to create the database by: determining whether the first token is included in the database; and upon determining the first token is not included in the database, adding the first token to the database.
4. The non-transitory, computer-readable storage medium of claim 1, comprising instructions to create the database by: obtaining a token from a trusted source; and determining whether the token is included in the database; and upon determining the token is not included in the database, adding the token to the database.
5. The non-transitory, computer-readable storage medium of claim 1, wherein instructions to compare the tokens comprise instructions to: compare the tokens using A-B matching, deterministic matching, or probabilistic matching.
6. The non-transitory, computer-readable storage medium of claim 1, comprising instructions to: obtain the first information; and obfuscate the first information by hashing and/or encryption to obtain the first information that is data masked.
7. A method comprising: obtaining a first token and a second token, wherein the first token includes first information that is data masked, wherein the second token includes second

information that is data masked, wherein the first information is different from the second information; comparing the first token and the second token to determine whether the first token and the second token match; upon determining that the first token and the second token do not match: obtaining a third token from a database storing unique tokens, wherein the third token includes third information that is data masked, wherein the third token is unique in the database; without accessing the first information in human-readable form and the third information in human-readable form, comparing the first token and the third token by comparing the first information that is data masked and the third information that is data masked to obtain a first indication of whether the first token and the third token match; without accessing the second information in human-readable form and the third information in human-readable form, comparing the second token and the third token by comparing the second information that is data masked and the third information that is data masked to obtain a second indication of whether the second token and the third token match; and upon determining that the first token and the third token match and that the second token and the third token match: determining that the first token and the second token match; and reducing memory footprint associated with the database by storing in the database a composite token including the first information and the second information.

8. The method of claim 7, comprising: creating a composite token including the first information and the second information, wherein the first information and the second information are data masked; and storing in the database an indication that the composite token is created.

9. The method of claim 7, comprising: creating a composite token including the first information and the second information, wherein the first information and the second information are data masked; determining whether the composite token is included in the database; and based on determining whether the composite token is not included in the database, selectively performing adding the composite token to the database, or increasing a confidence value associated with the composite token, wherein the confidence value indicates an accuracy associated with the composite token.

10. The method of claim 7, comprising creating the database by: determining whether the first token is included in the database; and upon determining the first token is not included in the database, adding the first token to the database.

11. The method of claim 7, comprising creating the database by: obtaining a token from a trusted source; and determining whether the token is included in the database; and upon determining the token is not included in the database, adding the token to the database.

12. The method of claim 7, wherein comparing the tokens comprises: comparing the tokens using A-B matching, deterministic matching, or probabilistic matching.

13. The method of claim 7, comprising: obtaining the first information; and obfuscating the first information by hashing and/or encryption to obtain the first information that is data masked.

14. A system comprising: at least one hardware processor; and at least one non-transitory memory storing instructions, which, when executed by the at least one hardware processor, cause the system to: obtain a first token and a second token, wherein the first token includes first information that is data masked, wherein the second token includes second information that is data masked, wherein the first information is different from the second information; compare the first token and the second token to determine whether the first token and the second token match; upon determining that the first token and the second token do not match: obtain a third token from a database storing unique tokens, wherein the third token includes third information that is data masked, wherein the third token is unique in the database; without accessing the first information in human-readable form and the third information in human-readable form, compare the first token and the third token by comparing the first information that is data masked and the third information that is data masked to obtain a first indication of whether the first token and the third token match; without accessing the second information in human-readable form and the third information in human-readable form, compare the second token and the third token by comparing the second information that is data

masked and the third information that is data masked to obtain a second indication of whether the second token and the third token match; upon determining that the first token and the third token match and that the second token and the third token match, reduce memory footprint associated with the database by storing in the database a composite token including the first information and the second information.

15. The system of claim 14, comprising instructions to: create a composite token including the first information and the second information, wherein the first information and the second information are data masked; and store in the database an indication that the composite token is created.

16. The system of claim 14, comprising instructions to: create a composite token including the first information and the second information, wherein the first information and the second information are data masked; determine whether the composite token is included in the database; upon determining that the composite token is not included in the database, add the composite token to the database; and upon determining that the composite token is included in the database, increase a confidence value associated with the composite token, wherein the confidence value indicates an accuracy associated with the composite token.

17. The system of claim 14, comprising instructions to create the database by: determining whether the first token is included in the database; and upon determining the first token is not included in the database, adding the first token to the database.

18. The system of claim 14, comprising instructions to create the database by: obtaining a token from a trusted source; determining whether the token is included in the database; and upon determining the token is not included in the database, adding the token to the database.

19. The system of claim 14, wherein instructions to compare the tokens comprise instructions to: compare the tokens using A-B matching, deterministic matching, or probabilistic matching.

20. The system of claim 14, comprising instructions to: obtain the first information; and obfuscate the first information by hashing and/or encryption to obtain the first information that is data masked.
